Automated Detection of Approval and Rejection Expressions

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***Abstract*—Depression is a critical global health issue, affecting millions with persistent emotional and cognitive challenges. Traditional diagnostic approaches often face limitations in scalability and accuracy, highlighting the need for advanced technological interventions. This study explores the application of machine learning models to detect approval and rejection expressions as indicators of emotional and psychological states. Using multiple data types, including facial expressions, body posture, and physiological signals, various ML techniques—such as Convolutional Neural Networks, Support Vector Machines, and ensemble learning methods—were applied to classify emotional responses. The dataset, collected through controlled testing sessions, was processed using a Python-based pipeline for automated feature extraction and classification. Results indicate that Random Forest and XGBoost models, particularly with Principal Component Analysis for dimensionality reduction, demonstrated the highest accuracy in distinguishing between approval and rejection expressions. The findings highlight the potential of AI-driven emotion recognition in mental health assessment.**

***Index Terms*—ML, classification, depression, study, emotion analysis**

I. Introduction

Depression is a significant global health challenge, impacting millions with symptoms such as persistent low mood, loss of interest, and cognitive impairment. While traditional therapeutic approaches, such as Cognitive Behavioral Therapy (CBT), have shown efficacy, they face limitations in scalability and response rates. The integration of artificial intelligence (AI) into mental health research presents new opportunities to analyze complex behavioral and physiological data, enabling more precise diagnosis and personalized interventions [1].

Machine Learning (ML) models have demonstrated remarkable success in image and video analysis. These models can effectively analyze data from various domains, including facial expressions and body posture, which are critical indicators of emotional and psychological states. By applying ML techniques to depression-related datasets, AI can detect patterns that may escape traditional analysis, providing deeper understanding and nuanced perspectives on the emotional and cognitive complexities of depression [2].

This work explores the use of ML models to analyze multiple data types data for detecting depression-related patterns. By examining facial expressions and kinesthetic markers, the research aims to enhance AI-driven approaches for mental health assessment. The findings contribute to the development of more effective and scalable diagnostic tools, offering valuable insights into the relationship between physiological indicators and psychological well-being [3].

II. Related Work

Recent advancements in deep learning have significantly improved emotion recognition, with CNNs playing a key role in capturing facial expression patterns. Instead of training models from scratch, researchers increasingly rely on transfer learning to leverage pre-trained networks, enhancing performance and efficiency across diverse datasets.

A. Transfer Learning for Emotion Recognition

Training CNNs from scratch often requires large-scale labeled datasets, which may not always be available. Transfer learning addresses this limitation by leveraging pre-trained models, such as VGG16 and ResNet, which have been trained on large datasets like ImageNet. These models retain learned low-level features while fine-tuning higher layers for task-specific adaptation [4], [5]. This technique improves generalization, reduces computational costs, and mitigates overfitting, making it particularly useful for small datasets [6].

B. Datasets for Pretraining and Fine-tuning

To enhance model performance, studies have utilized large-scale datasets for pretraining before fine-tuning on emotion-specific tasks. The UCF101 dataset, originally designed for human action recognition, provides diverse video-based data and serves as an effective pretraining resource. The Kaggle Emotion Detection (DeepFace) dataset, containing annotated facial expression images, is commonly used for fine-tuning models to detect emotion-specific patterns [7].

C. Challenges in Emotion Recognition

Despite its advantages, CNN-based emotion recognition faces challenges such as data imbalance, computational inefficiency, and sensitivity to variations in lighting and occlusions. Data augmentation techniques, including geometric transformations and normalization, have been widely employed to improve model robustness [2], [4]. Additionally, hybrid models integrating CNNs with Long Short-Term Memory (LSTM) networks or employing 3D CNNs have shown promise in

capturing spatiotemporal dependencies in dynamic facial expressions [8].

III. Methodology

This project employs machine learning techniques to analyze physiological and behavioral data from the Empkins D02 dataset [1]. A Python-based preprocessing pipeline was developed to automate data extraction, filtering, and feature computation. The extracted features, including statistical measures, were categorized into approval and rejection classes, forming the foundation for machine learning-based classification. Various models were later applied to evaluate the predictive performance, ensuring a robust analysis of the dataset.

IV. Detecting Expressions using Machine Learning

A. Testing and Data Collection

I actively participated in running 19 testing sessions aimed at collecting data for the diagnosis of depression, working in collaboration with psychologists. The data collection process was conducted in a controlled lab environment using a comprehensive set of advanced tools. A Sony camera and Azure Kinect camera were employed to capture high-quality frontal and depth video recordings, respectively, while a microphone recorded audio for backup purposes. Two radar systems, operating at 24 GHz and 61 GHz, were used for contactless monitoring of breathing and heart rate. Additionally, a Biopac Acknowledge system was utilized to measure psychophysiological signals such as heart rate, respiration, and facial EMG.

Participants interacted with the EmpkinS app on a smartphone during the sessions, while I ensured the seamless operation of all devices, monitored the data streams, and verified signal quality in real time. My responsibilities included setting up and troubleshooting equipment, coordinating with psychologists to maintain testing protocols, and managing data storage and transfer. This process ensured the collection of high-quality multi-modal data for machine learning analysis and clinical research.

B. Dataset Description

The dataset was gathered during a psychological study of depression patients. It is structured across multiple directories, including raw data and processed video-based data. Data from these sources were extracted in CSV format for analysis, focusing on variables relevant to movement and behavioral patterns. The raw data repository contains sensor recordings and experimental metadata, while the processed video data provides extracted variables linked to specific tasks and participant responses. These datasets collectively enable a comprehensive analysis of physiological responses under different experimental conditions.

C. Data Processing and Feature Extraction

To automate data extraction and preprocessing, a Python-based tool was implemented. This tool systematically imports, filters, and organizes sensor data collected from all participants. It scans the dataset directories, locates relevant files while ignoring unnecessary ones, and processes only specified patient numbers. Pattern matching is used to identify task-related filenames, ensuring that only the required data is extracted for analysis.

The algorithm iterates over all timesheet CSV files, which contain information linking specific timeframes of sensor data to corresponding patient tests. Task labels such as ”coping_ng” (rejection) and ”training_pg” (approval) are used to classify the data. The processing function ensures that only relevant tasks and timestamps within a predefined range are considered. Time differences are computed in milliseconds to accurately retain only the necessary rows for analysis.

For each identified CSV file, the script reads the data, verifies the presence of a timestamp column, and filters rows based on the computed time range. The filtered data is then stored in global DataFrames according to the respective task classification. Once all relevant data is collected, the script calculates statistical features, including the mean and standard deviation for both approval and rejection datasets, summarizing key patterns for further analysis.

This workflow is fully automated, iterating through patient records, identifying relevant tasks, and processing data accordingly. After execution, summaries of the approval and rejection datasets are generated, providing a well-organized foundation for machine learning analysis. The generated dataset contains a total number of 34 participants, each providing data for ”approval” and ”rejection”, resulting in a combined dataset of 68 entries.

D. Machine Learning Models and Evaluation

In this phase, I implemented multiple machine learning models to classify the data into approval and rejection categories. The dataset was split into 80% training and 20% testing sets to evaluate model performance effectively. I utilized the scikit-learn library to implement and train the models. The primary evaluation metrics used were Accuracy, Precision, Recall, and F1-score to assess the classification performance comprehensively.

The models implemented include K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naive Bayes, Random Forest, and XGBoost. For model evaluation, I experimented with hyperparameter tuning methods such as GridSearchCV and RandomizedSearchCV. For RandomizedSearchCV, I tested 300 different hyperparameter settings per model to optimize performance. In the default (non-optimized) setting. The evaluation of each model was based on its ability to classify the given dataset correctly while handling potential class imbalances.

The classification function was structured to select and train a model based on user input. For each model, I assessed the model’s performance using the test set, computing evaluation

metrics, and printing the results. I also explored different optimization strategies to enhance model performance.

1) **K-Nearest Neighbors (KNN)**
KNN is a simple and intuitive classification algorithm that assigns a class label based on the majority vote of the nearest neighbors in the feature space. KNN does not require a training phase and works well for small to medium-sized datasets, though it can be computationally expensive for large datasets. Dimensionality reduction via PCA can be applied to enhance its performance, particularly in high-dimensional data [9].
2) **Support Vector Machine (SVM)**
SVM aims to find an optimal hyperplane that maximizes the margin between different classes. SVM is effective in high-dimensional spaces and can be used for both linear and non-linear classification tasks. SVM's performance can be further improved with appropriate kernel selection and regularization parameters, and dimensionality reduction through PCA [9].
3) **Naive Bayes**
Naive Bayes is a probabilistic classifier that applies Bayes' Theorem under the assumption that features are conditionally independent given the class label. It is particularly effective for classification tasks involving high-dimensional and sparse data and is computationally efficient. The Gaussian Naive Bayes variant was used to model the approval and rejection tasks. However, for optimal accuracy, the data should be approximately Gaussian distributed [9].
4) **Random Forest**
Random Forest, an ensemble learning method, constructs a multitude of decision trees and aggregates their predictions to enhance classification accuracy. Random Forest helps mitigate overfitting by averaging the predictions from various trees, making it a robust model for complex datasets. In this case, it performed well in distinguishing between the approval and rejection classes [9].
5) **XGBoost**
XGBoost is a gradient boosting framework that iteratively improves the model by correcting the errors of previous trees. It is known for its efficiency and effectiveness in competitive machine learning tasks. XGBoost's performance was optimized through hyperparameter tuning and dimensionality reduction via PCA, yielding high accuracy in classifying approval and rejection tasks [9].

E. Dimensionality Reduction using PCA

To optimize the computational efficiency and improve the generalization of the models, I applied Principal Component Analysis (PCA) as a dimensionality reduction technique. PCA helps to transform high-dimensional data into a lower-dimensional space while preserving essential variance.

The PCA implementation involved standardizing the dataset using StandardScaler to ensure that all features contribute equally to the principal components. Different numbers of principal components were tested, ranging from 0 (no PCA) to 68 components, to determine the optimal balance between dimensionality reduction and information retention. The maximum number of components is limited by the smaller of the number of features and number of sample of the dataset.

The principal components were computed using the 'full' SVD solver, which ensures accurate decomposition of the feature space. The transformed dataset was then used to train the machine learning models, allowing to compare their performance with and without dimensionality reduction. This process helped evaluate the impact of PCA on classification accuracy and computational efficiency.

V. Result

In the course of this project, data from a psychological experiment was collected to analyze emotional responses in patients with depression. A Python-based tool was developed to automate the import, filtering, and organization of raw sensor data into a structured dataset suitable for training machine learning models. The dataset was categorized into "approval" and "rejection" classes, providing both raw sensor readings and statistical features such as mean and standard deviation for each parameter. The processed data was stored in DataFrames and exported as CSV files for further machine learning applications.

Various machine learning classifiers, KNN, SVM, Naive Bayes, Random Forest, and XGBoost, were trained and tested to create a binary classifier distinguishing between "approval" and "rejection." Each model was evaluated using default configurations and further optimized through hyperparameter tuning with both grid search and random search. Since grid search yielded suboptimal results, random search was preferred for model optimization.

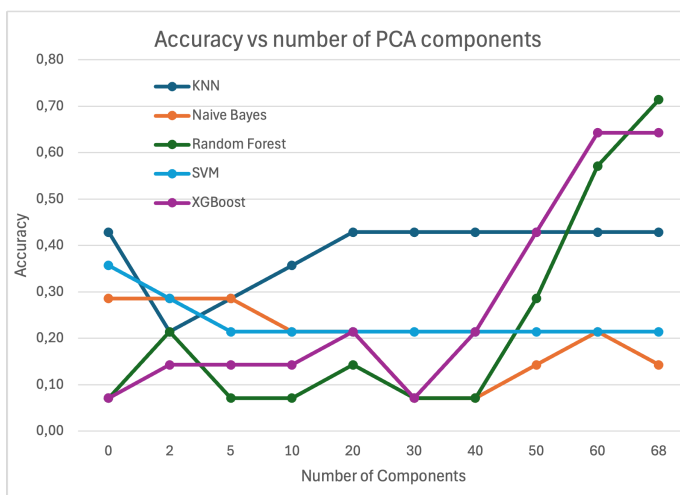

Fig. 1. Accuracy vs number of PCA components for various machine learning models used in binary classification in 'approval' and 'rejection' tasks, without hyperparameter optimization.

The dataset consists of 1199 features, with the selection of relevant features varying based on the type of machine

learning model employed. Feature importance is determined dynamically depending on the specific algorithm used for classification. The most relevant features for the models are illustrated in Figure 2 for random forest and in Figure 3 for XGBoost.

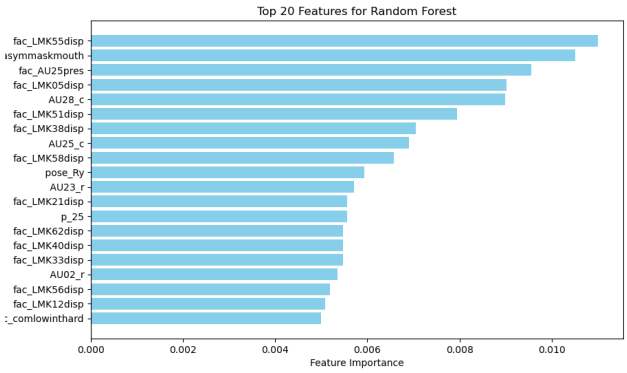

Fig. 2. Main features of random forest model without hyperparameter optimization or PCA.

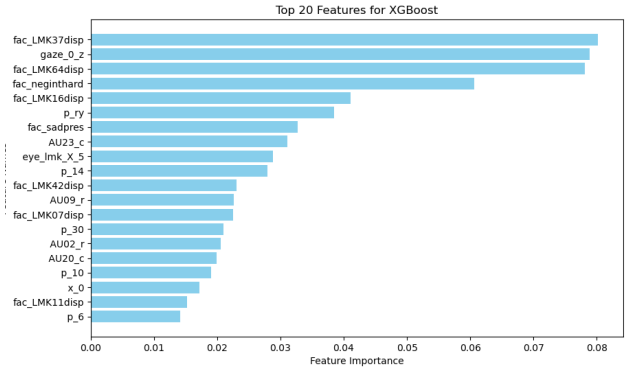

Fig. 3. Main features of XGBoost model without hyperparameter optimization or PCA.

To enhance classification accuracy, PCA was applied to reduce dimensionality, and multiple numbers of components were tested. Figure 1 illustrates the accuracy of the models across different PCA component selections. The results indicate a general trend of improved accuracy with an increasing number of PCA components, with Random Forest and XGBoost demonstrating particularly strong improvements. The highest recorded accuracy was 71% for the Random Forest model with 68 PCA components, while XGBoost achieved a maximum accuracy of 63%. Further accuracy improvements were constrained by the dataset's limited sample size, restricting the maximum PCA features to 68.

To explore potential improvements, hyperparameter optimization using random search was conducted, as shown in 4. Randomized search was used with 300 iterations of hyperparameter optimization per model to explore optimal

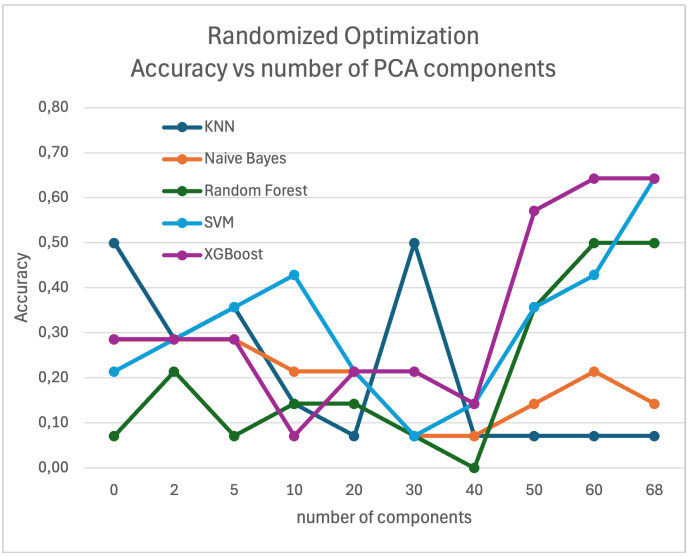

Fig. 4. Accuracy vs number of PCA components for various machine learning models used in binary classification of 'approval' and 'rejection' tasks, with random hyperparameter optimization.

configurations. The best default and best performing hyperparameter settings found with randomized search are shown in Figure 5. Compared to the non-optimized models, the results exhibited higher accuracy variation due to the stochastic nature of random search. However, the overall trend of increasing accuracy with more PCA components remained. Despite some fluctuations, random search did not significantly enhance model performance for this dataset. The best performing model with and without optimization is with 68 PCA components, respectively.

Random Forest	Default	Randomized Search
n_estimators	100	100
max_depth	None	30
min_samples_split	2	5
min_samples_leaf	1	1
max_features	sqrt	log2
bootstrap	True	True

XGBoost	Default	Randomized Search
n_estimators	100	100
max_depth	6	3
min_child_weight	1	5
learning_rate	0.3	0.1

Fig. 5. Hyperparameters of best performing random forest and XGBoost models with 68 PCA components using default settings and randomized search, respectively.

VI. Conclusion

This project demonstrated the potential of machine learning models in detecting depression-related emotional responses through physiological and behavioral data analysis. While Random Forest and XGBoost showed the highest accuracy, improvements were constrained by the dataset's limited size. Although PCA effectively enhanced classification performance, hyperparameter optimization via random search did not yield significant additional gains. Future work should focus on refining the machine learning models for improved classification accuracy and potentially explore deep learning techniques to improve model generalization and accuracy.

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